Homework 2 - EE4520 Analog CMOS Design 1

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Contents

1 Deliverable 1 - Table of Results

#	Design Item	Unit	Spec	Achieved
1	Design Number	[-]	27	27
2	SNR	[dB]	80.64	80.85
3	A_{settle}	[dB]	63	63.32
4	T_{settle}	[ns]	2800	2800
5	BW_{cl} from open-loop AC sims	[MHz]	—	0.329
6	BW_{cl} from closed-loop AC sims	[MHz]	—	0.329
7	Time-constant τ_{cl} from closed-loop AC sims	[ns]	—	483.1
8	$T_{40\text{dB}}$ (time to reach accuracy of 40 dB)	[ns]	—	1396.9
9	$T_{48.69\text{dB}}$ (time to reach accuracy of 48.69 dB)	[ns]	—	1894.3
10	Time-constant τ_{cl} from transient sim	[ns]	—	497.4
11	Power dissipation	[mW]	Lowest possible	0.1119
12	$I(V_{dd})$	[mA]	Lowest possible	0.0622
13	Total integrated noise	[µV]	—	76.93
14	Input step voltage	[mV]	150	150
15	Output step voltage	[V]	1.2	1.2
16	C_{in}	[pF]	—	18
17	C_{fb}	[pF]	—	2.25
18	C_{load}	[pF]	—	18
19	C_{CM}	[pF]	—	2.25
20	R_{big}	[GΩ]	Big	100
21	C_{big}	[nF]	Big	1000
22	Bonus points on FoM_{dB}	[-]	0	0
23	$\text{FoM}_{lin} = 2\pi \cdot \text{Power} \cdot T_{8.6\text{dB}}/\text{SNR}^2_{lin}$	[J]	$\leq$ 3.98e−18	3.02e−18
24	$\text{FoM}_{dB} = -10\log_{10}(\text{FoM}_{lin})$	[dB]	$\geq$ 174.00	175.20

Computation notes

- BW_{cl} (open-loop): frequency where $|A\beta| = 1 \Rightarrow 0.329$ MHz.
- BW_{cl} (closed-loop): -3 dB frequency of closed-loop gain $\Rightarrow 0.329$ MHz.
- τ_{cl} (AC): $\tau_{cl} = \dfrac{1}{2\pi BW_{cl}} = \dfrac{1}{2\pi \cdot 329\text{kHz}} = 483.1$ ns.
- τ_{cl} (transient): $\tau_{cl} = T_{48.69\text{dB}} - T_{40\text{dB}} = 1894.3 - 1396.9 = 497.4$ ns.
- SNR: $\text{SNR}_{dB} = 20\log_{10}\left(\dfrac{0.8485}{76.93 \cdot 10^{-6}}\right) = 80.85$ dB.
- Power: $P = \overline{I(V_{dd})} \cdot 1.8\text{ V} = 62.2\ \mu\text{A} \cdot 1.8\text{ V} = 0.1119$ mW.

Design parameters

ScaleA = 0.25, Mn34 = 200, C_{in} = 18 pF, I_{bm} = 350 μA, W = 1 μm, L = 0.18 μm, Mp = 60, Mn = 11, mp_ratio = 5.45, Rn = 5.8, Rp = 7.5, Aadd = 100.

2 Deliverable 2 - Schematic with All Node Voltages

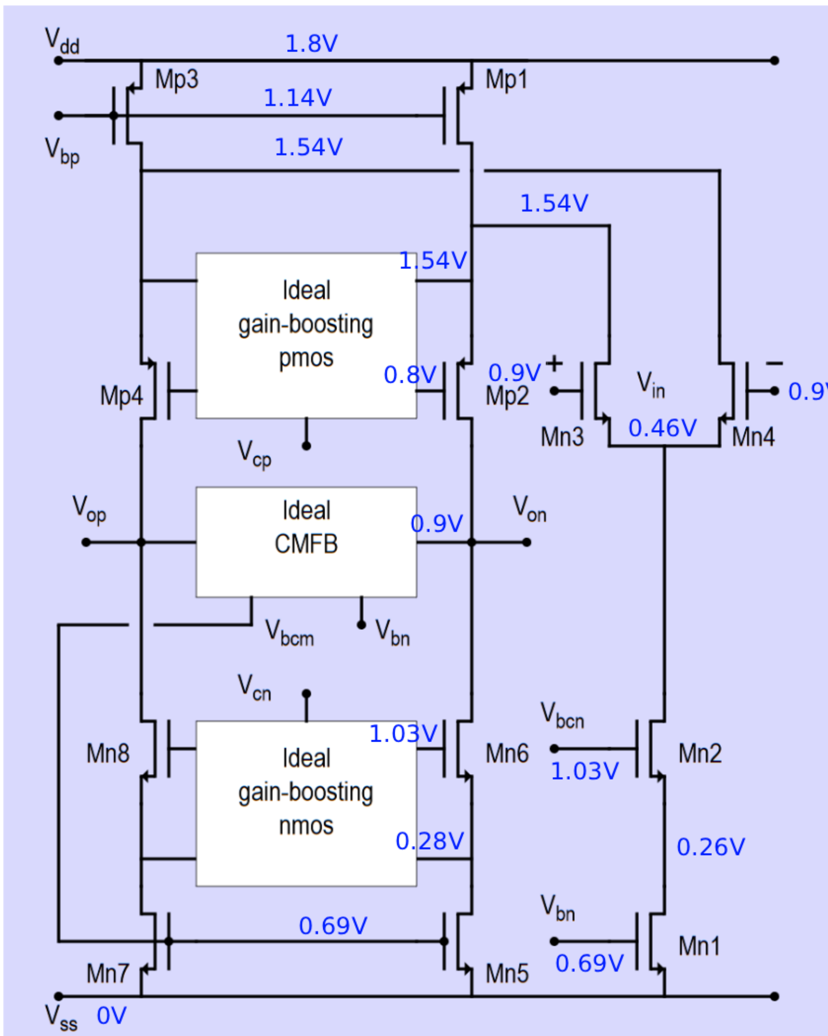

Figure 1: Folded-cascode OpAmp with ideal gain-boosting and annotated DC node voltages.

3 Deliverable 3 - Schematic with All Branch Currents

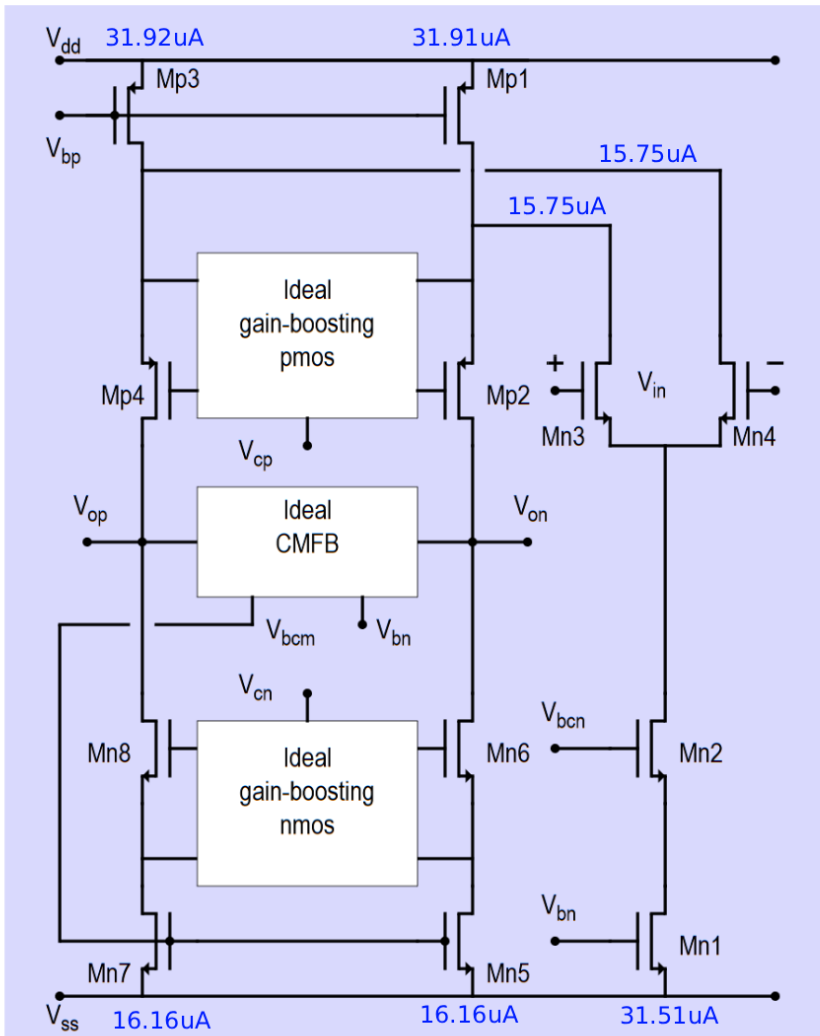

Figure 2: Folded-cascode OpAmp with annotated DC branch currents.

4 Deliverable 4 - Bode Diagram of A and $A\beta$ (Gain and Phase)

4.1 Unloaded open-loop simulation

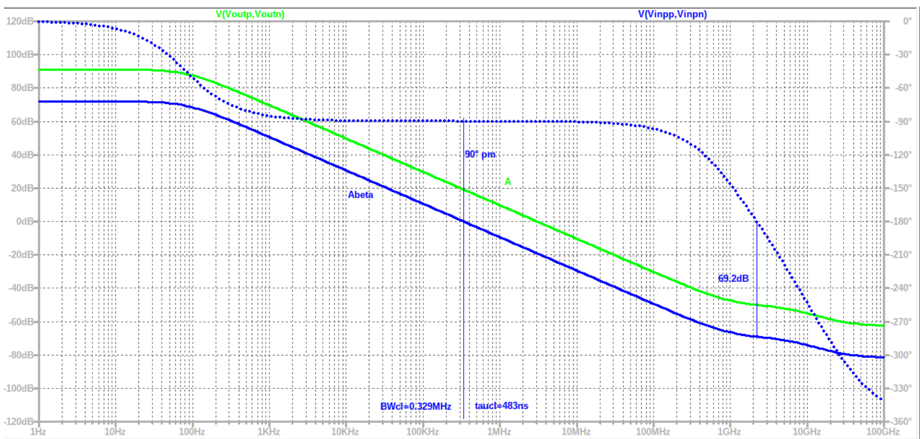

Figure 3: Bode diagram of open-loop gain A and loop gain $A\beta$ (unloaded).

Key extracted metrics:

- Closed-loop bandwidth from unity loop-gain crossing: $BW_{cl} = 0.329$ MHz.
- Closed-loop time-constant: $\tau_{cl} = \dfrac{1}{2\pi BW_{cl}} = 483.1$ ns.
- Low-frequency separation between A and $A\beta$ equals $20\log_{10}(8) = 18.06$ dB, consistent with $A_{cl} = 8$.

4.2 Loaded open-loop simulation (with replica loading)

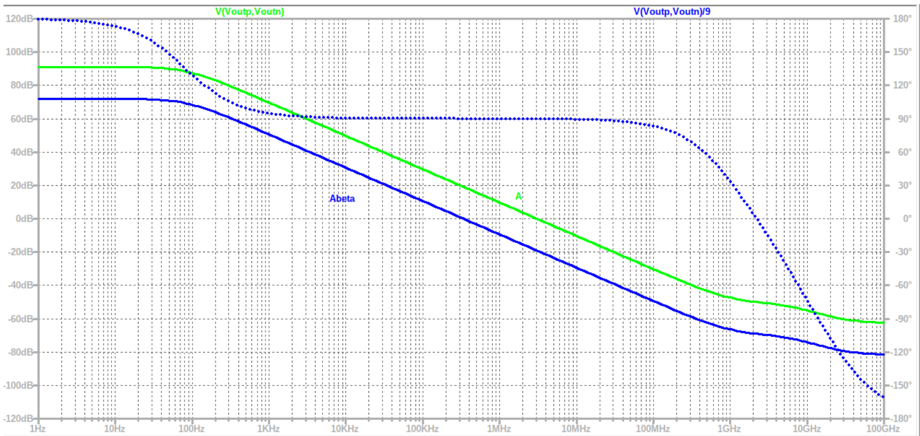

Figure 4: Bode diagram of open-loop gain A and loop gain $A\beta$ (loaded).

The loaded and unloaded cases yield essentially the same BW_{cl} and stable behavior (high phase margin), indicating limited sensitivity to the applied replica loading for this configuration.

5 Deliverable 5 - Bode Diagram of Closed-Loop Gain (Gain and Phase)

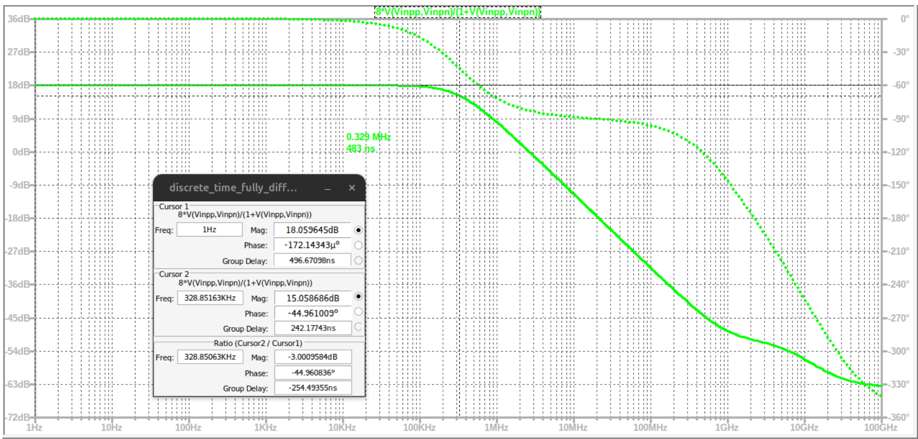

Figure 5: Closed-loop transfer function magnitude and phase.

From the plot:

- Passband gain: $20 \log_{10}(8) = 18.06$ dB.
- -3 dB bandwidth: $BW_{cl} \approx 0.329$ MHz.
- Corresponding time constant: $\tau_{cl} = \dfrac{1}{2\pi BW_{cl}} = 483.1$ ns.

The BW_{cl} obtained here matches the unity-crossing of $|A\beta|$ from Deliverable 4, as expected for a predominantly first-order closed-loop system.

6 Deliverable 6 - Settling Accuracy Versus Time

The settling accuracy is evaluated using:

$$\text{Accuracy}_{dB}(t) = 20\log_{10}\left(\frac{1.2}{|V(\text{virtualground})| + 1\text{n}}\right).$$

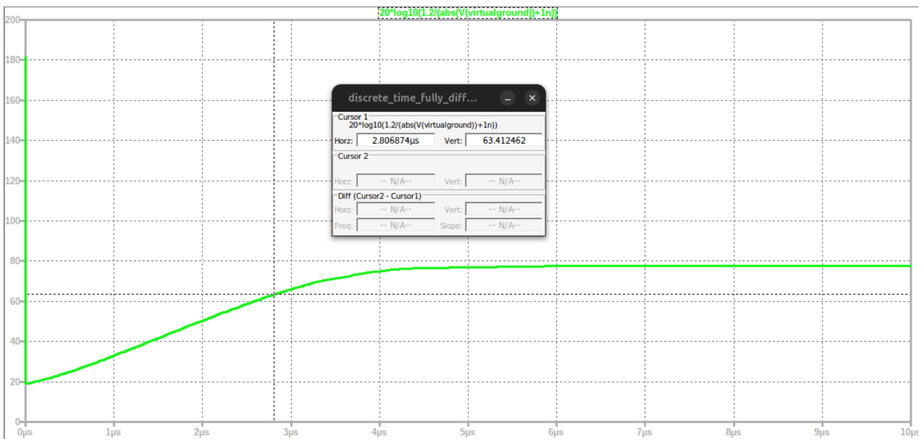

Figure 6: Settling accuracy versus time.

At $T_{\text{settle}} = 2.8\ \mu$s, the achieved accuracy is ≈ 63.3 dB, meeting the 63 dB target.

7 Deliverable 7 - Settling Accuracy with T_{settle}, $T_{40\text{dB}}$, and $T_{48.69\text{dB}}$

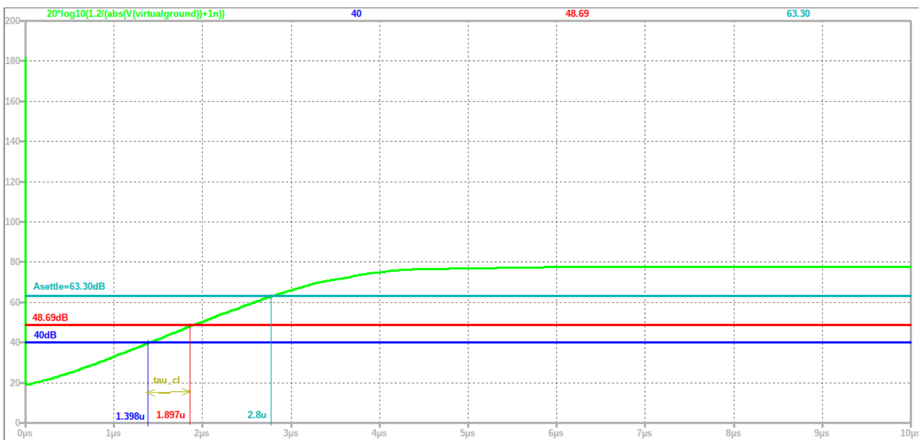

Figure 7: Settling accuracy with timing markers for time-constant estimation.

Extracted timing points:

- $T_{40\text{dB}} = 1396.9$ ns,
- $T_{48.69\text{dB}} = 1894.3$ ns,
- $\tau_{cl} = T_{48.69\text{dB}} - T_{40\text{dB}} = 497.4$ ns.

Comparison of τ_{cl}:

Source	τ_{cl} [ns]
From AC ($BW_{cl} = 0.329$ MHz)	483.1
From transient ($T_{48.69\text{dB}} - T_{40\text{dB}}$)	497.4

8 Deliverable 8 - Output Noise Power Density Versus Frequency

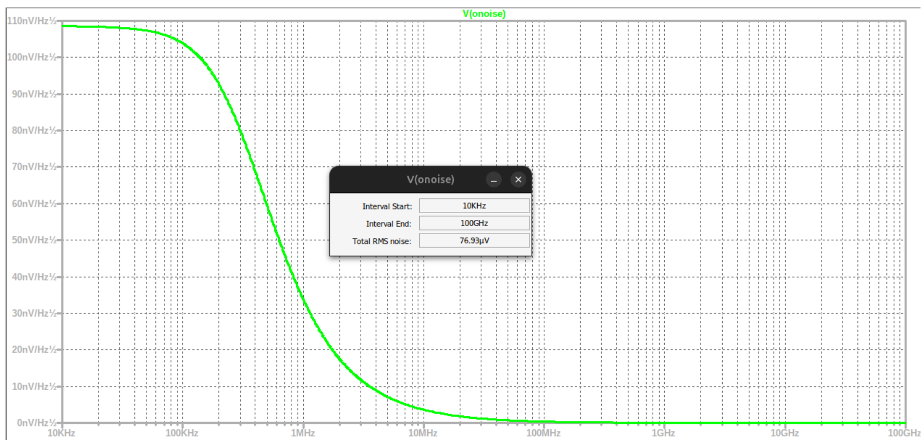

Figure 8: Output noise density versus frequency and integrated RMS noise (10 kHz – 100 GHz).

Integrated output RMS noise (10 kHz to 100 GHz): 76.93 μV_{rms}, yielding:

$$\mathrm{SNR}_{dB} = 20 \log_{10}\left(\frac{0.8485}{76.93 \cdot 10^{-6}}\right) = 80.85 \text{ dB}.$$

9 Deliverable 9 - Design Process Description

9.1 Design constraints and dominant trade-offs

This design is primarily noise-limited: the SNR requirement of 80.64 dB constrains the integrated output noise to $\leq 78.8\ \mu\text{V}_{rms}$, while the settling target ($T_{\text{settle}} = 2.8\ \mu$s) sets the required closed-loop bandwidth. Key observations:

- Increasing capacitor values reduces integrated noise (via kT/C) but increases required charge transfer, thus requiring higher g_m (more current) to maintain settling.
- The input pair (MN3/MN4) dominates thermal noise; biasing it in weak/moderate inversion maximizes g_m/I_D and improves noise efficiency.
- Current-source devices (MN1, MN5–8, MP1–4) contribute noise; biasing them closer to strong inversion reduces their relative contribution while maintaining headroom.
- Global scaling (ScaleA) increases all currents, improving speed/noise but increasing power roughly proportionally.
- Therefore, C_{in} and input-pair sizing are the main levers to meet SNR at minimum power.

9.2 Design steps

1. **Capacitor selection:** Choose $C_{in} = 18$ pF with $C_{fb} = C_{in}/8 = 2.25$ pF, $C_{load} = C_{in}$, and $C_{CM} = C_{fb}$ to achieve 76.93 μV_{rms} noise.
2. **Input pair sizing:** Use large multiplicity (Mn34 = 200) to increase effective g_m at low current and reduce dominant noise.
3. **Power minimization:** Set ScaleA = 0.25 and $I_{bm} = 350\ \mu$A, yielding $\overline{I(V_{dd})} \approx 62.2\ \mu$A and $P \approx 0.1119$ mW. Sweeps of I_{bm} and C_{in} identify the minimum-power solution.

9.3 Link between open-loop, closed-loop, and settling results

The unity crossing of $|A\beta|$ gives:

$$BW_{cl} \approx 0.329 \text{ MHz}, \quad \tau_{cl,\text{AC}} = \frac{1}{2\pi BW_{cl}} = 483.1 \text{ ns}.$$

The separation $20\log_{10}(8) \approx 18.06$ dB confirms $A_{cl} = 8$, and the closed-loop AC simulation gives the same BW_{cl}.

From transient:

$$\tau_{cl,\text{tran}} = T_{48.69\text{dB}} - T_{40\text{dB}} = 497.4 \text{ ns},$$

which is $\approx 3\%$ higher than $\tau_{cl,\text{AC}}$. At 2.8 μs, $A_{\text{settle}} \approx 63.3$ dB, SNR = 80.85 dB, and $\text{FoM}_{\text{dB}} = 175.20$ dB.

A Appendix - Simulation Schematics

A.1 Reference Schematic with Voltages and Currents

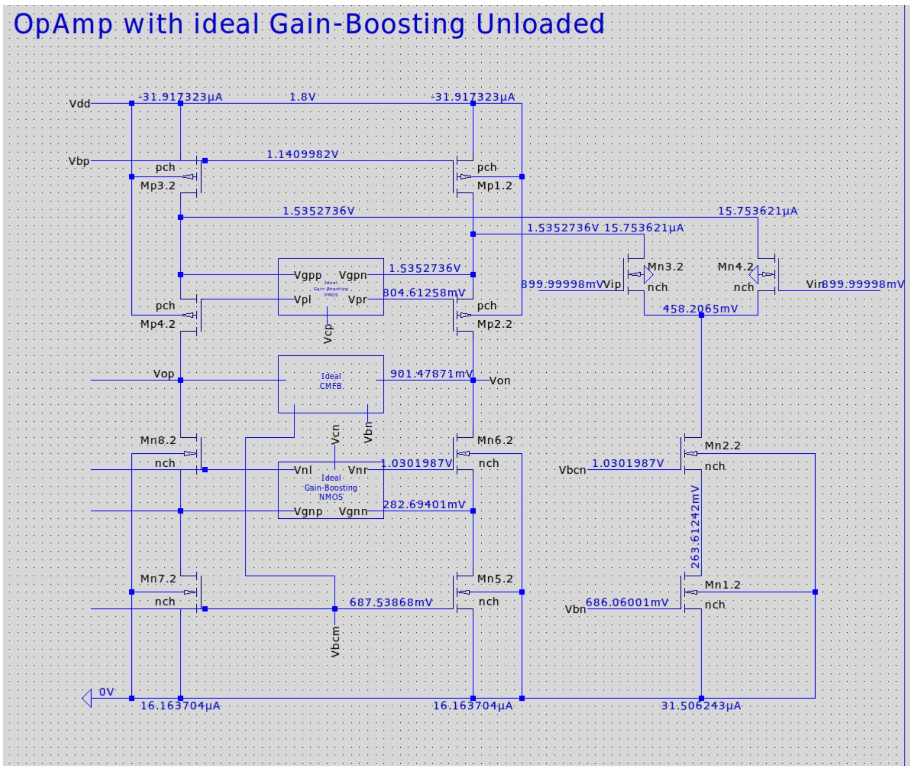

Figure 9: Complete OpAmp schematic with annotated node voltages and branch currents.

A.2 Open-Loop AC Simulation (Unloaded)

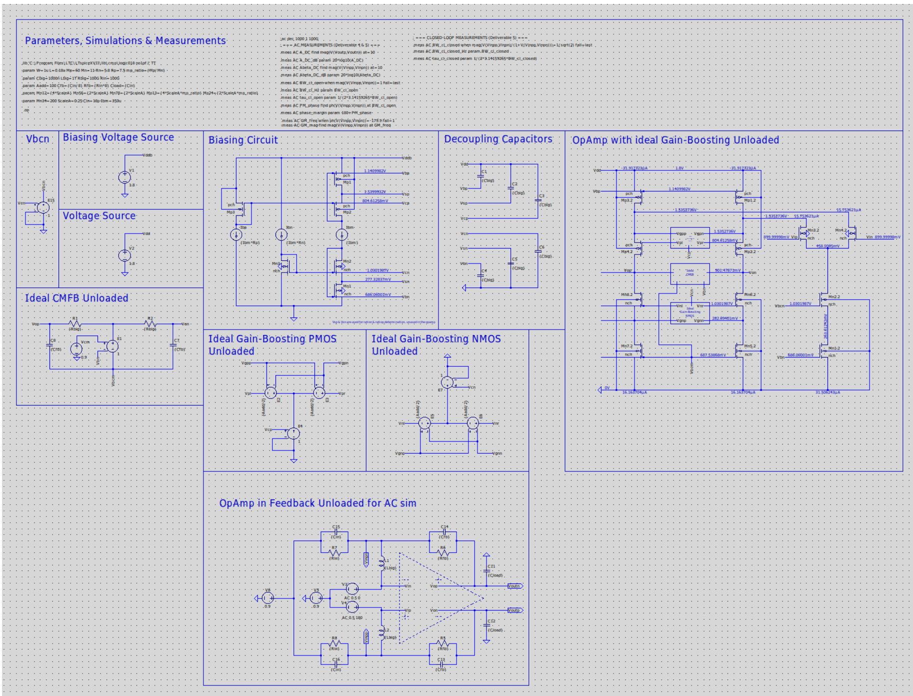

Figure 10: Schematic used for open-loop AC simulation without replica loading.

A.3 Open-Loop AC Simulation (Loaded)

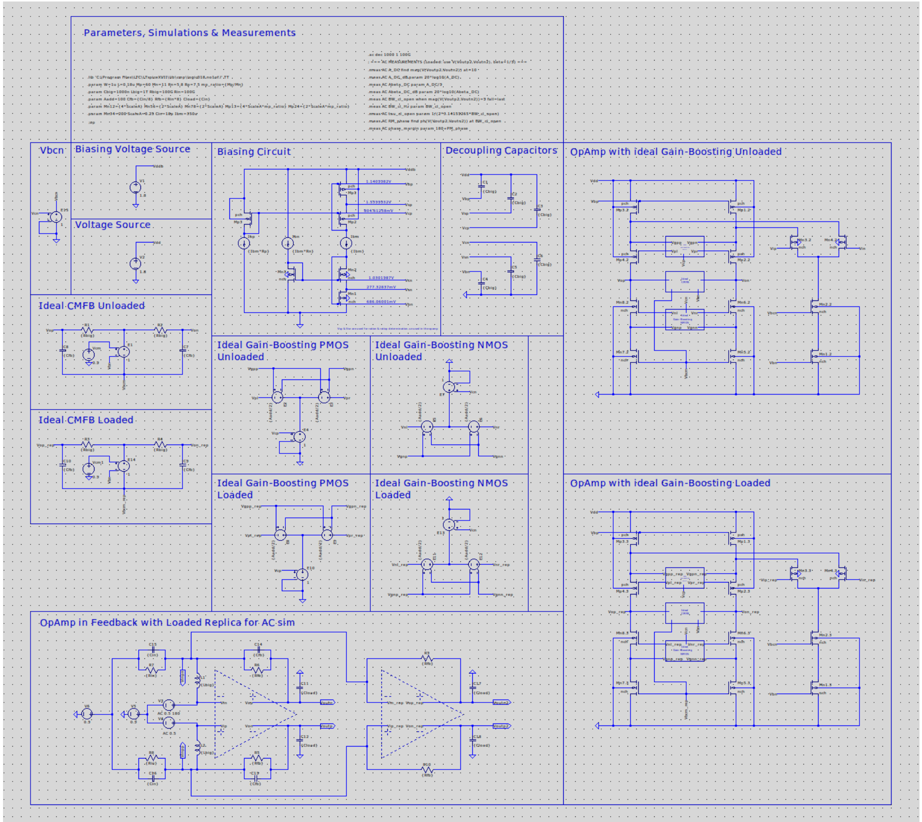

Figure 11: Schematic used for open-loop AC simulation with replica loading.

A.4 Closed-Loop Noise Simulation

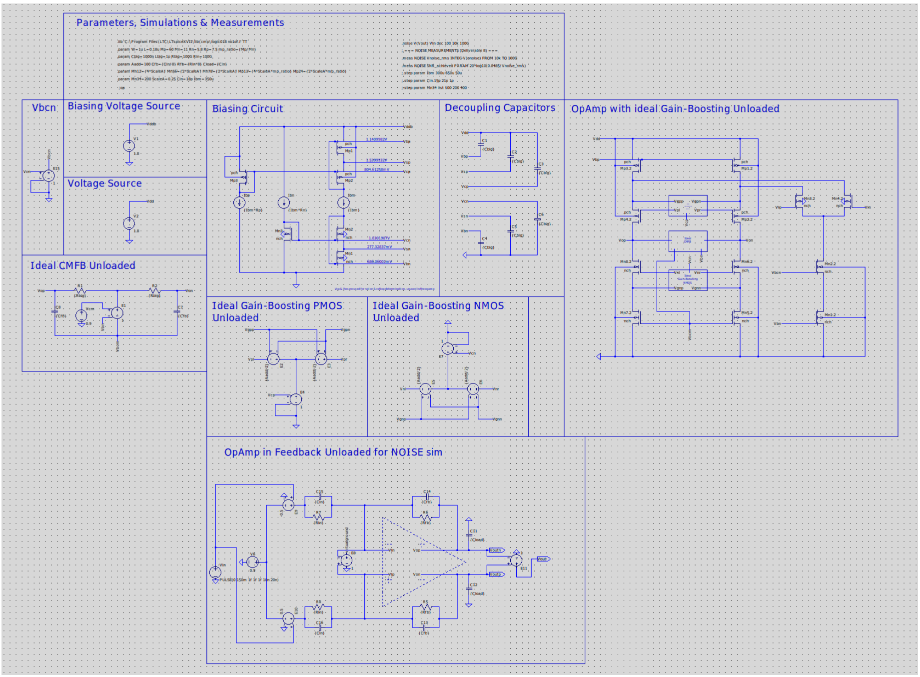

Figure 12: Schematic used for output noise simulation in closed-loop configuration.

A.5 Closed-Loop Transient Simulation

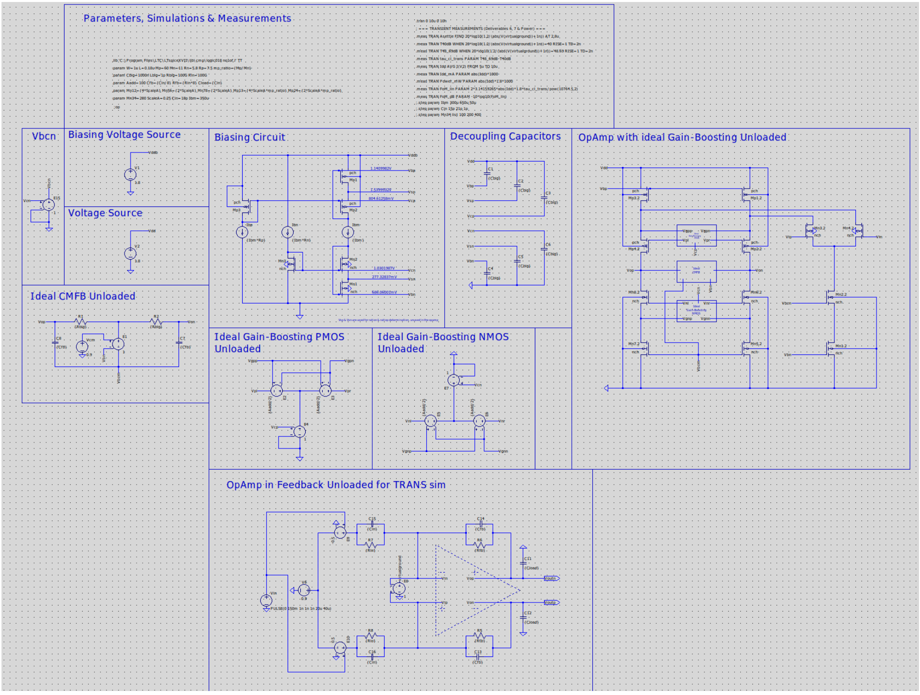

Figure 13: Schematic used for transient settling simulations.